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import socket

# Forward DNS Lookup (Domain → IP)
def forward_lookup(domain):
    try:
        ip = socket.gethostbyname(domain)
        print(f"IP Address of {domain}: {ip}")
    except socket.gaierror:
        print("Host not found.")

# Reverse DNS Lookup (IP → Domain)
def reverse_lookup(ip_address):
    try:
        host = socket.gethostbyaddr(ip_address)
        print(f"Host Name of {ip_address}: {host[0]}")
    except socket.herror:
        print("Host not found.")

def main():
    print("DNS Lookup Program")
    print("1. Forward Lookup (Domain → IP)")
    print("2. Reverse Lookup (IP → Domain)")

    choice = input("Enter your choice (1/2): ")

    if choice == '1':
        domain = input("Enter Domain Name: ")
        forward_lookup(domain)

    elif choice == '2':
        ip_address = input("Enter IP Address: ")
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reverse_lookup(ip_address)
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else:
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    print("Invalid choice. Exiting.")
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if __name__ == "__main__":
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    main()
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